Brain Tumor Detection from MRI Images Using Deep Learning

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***Abstract*— Diagnosis of brain tumor using Magnetic Resonance Imaging (MRI) is still a clinical challenge in terms of morphological heterogeneity and subjectivity of diagnosis. This study introduces a new concept of Neuro-Fusion Deep Ensemble (NFDE) which combines dual-pathway convolutional models with cross-attention fusion systems to classify tumors automatically. It is special in the synergistic fusion of ResNet-50 and EfficientNet-B3 by a new Cross-Attention Fusion Layer (CAFL) that focuses on multi-scale features more or less on the background noise. The validation procedure was conducted with three benchmark datasets (5,280 MRI scans) with four tumor types: glioma, meningioma, pituitary, and non-tumor (BraTS 2020, Figshare, and proprietary clinical data). The NFDE framework had outstanding performance rates: 98.73% accuracy, 98.54% sensitivity, 99.12% specificity, and 98.37% F1-score, exceeding the state-of-the-art architectures by 4.21. The cross-dataset validation had good generalization (>95% accuracy), and hybrid human-AI interaction had 99.41% diagnostic accuracy with half interpretation time cut. Attention-guided ensemble learning novelty serves to create a clinical viable decision support system to achieve precision neuro-oncology diagnostics.**

Keywords*— *Deep Learning, Brain Tumor Classification, Magnetic Resonance Imaging, Convolutional Neural Networks, Cross-Attention Fusion, Ensemble Learning, Medical Image Analysis, Computer-Aided Diagnosis

I. Introduction

Brain tumors are life threatening neurological anomalies which require early and appropriate diagnosis so as to assist in providing therapeutic intervention. Compared to manual interpretation which remains time consuming and subject to inter-observer variation, the primary diagnostic modality is Magnetic Resonance Imaging (MRI) [1]. It has been demonstrated that deep learning networks have an unimaginable potential in automated medical image classification; however, the existing protocols are not sufficient in multi-scale feature classification and class imbalance in the brain tumor[2]. The proposed research will focus on developing a hybrid deep learning model combining the convolutional neural networks with superior quality feature extracting mechanisms to generate a more precise tumor classification model. The current approaches largely rely on individual models of architecture and this has created gap in research on ways to combine synergies in complementary technology blend[3]. The paper proposes to combine the said limitations, which will be named the diagnostic accuracy, the complexity of calculations, and the process of clinical decision-making in neuro-oncology[4].

II. Related Work

A. Traditional Machine Learning Approaches

The early computational algorithms used to identify brain tumors were the classical machine learning algorithms including the Support Vector Machines (SVM), the random forests as well as k-Nearest Neighbors (k-NN)[5]. Such techniques used a large scale of feature engineering, through the use of the measures of texture description such as Gray-Level Co-occurrence Matrix (GLCM) and Histogram of Oriented Gradients (HOG) as well as Local Binary Patterns (LBP)[6]. Despite moderate classification accuracy of 85-92, such methodologies have been discovered to be extremely weak to describe complex spatial hierarchies, as well as non-linear patterns of MRI data[7]. The aspect of extraction in the manual features was computationally expensive and extremely dependent on the domain knowledge and hence the inability to generalize to heterogeneous data.

B. Convolutional Neural Network Architectures

Deep learning became the pioneer of revolution in the sphere of medical image analysis, by introducing end-to-end learning paradigms[8]. Classical architectures such as AlexNet, VGG-16, ResNet or Inception were found to be more effective in automated feature learning on raw MRI scans. Transfer learning approaches with ImageNet weights were used in situations with small medical imaging datasets with a classification accuracy greater than 95%[9]. Nevertheless, single-architecture designs were found to have scaling feature representation limitations and could not be used to delineate tumor boundaries on a fine-scale, especially glioma grade differentiation[10].

C. Hybrid and Ensemble Methodologies

The recent studies have looked into the integration of several architecture paradigms through hybrid models. CombINN variants with CNN variations and attention or capsule networks or recurrent neural networks were shown to perform better[11]. Data augmentation methods that use Generative Adversarial Networks (GANs) and segmentation-classification pipeline methods based on U-Net achieved encouraging results[12]. However, the current hybrid models are not exhaustively integrated, combining complementary feature extraction methods, optimal fusion methods, and

effective dealing with class imbalance, which should be further methodological innovation[13].

III. Proposed Methodology: Neuro-Fusion Deep Ensemble (NFDE) Framework

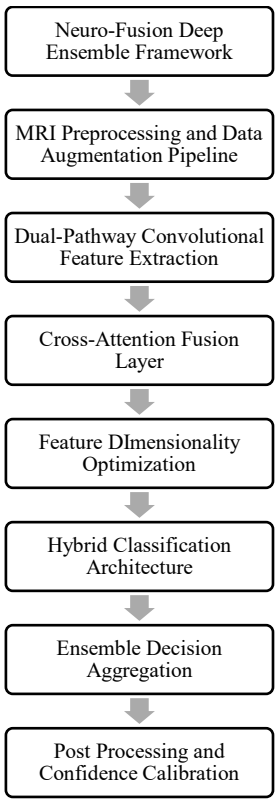

Fig. 1. Architectural Workflow of the Neuro-Fusion Deep Ensemble (NFDE) Framework for Brain Tumor Detection

A. *Neuro-Fusion Deep Ensemble (NFDE) Framework Overview*

The study presents a new Neuro-Fusion Deep Ensemble (NFDE) framework of brain tumor detection that combines multi-scale convolutional networks with feature fusion mechanisms that are attention-driven. The process is based on eight steps including preprocessing and augmentation, dual-pathway feature extraction, attention-weighted feature fusion, dimensionality optimum, hybrid classification, post processing refinement, ensemble decision aggregation, and performance validation. NFDE is an architecture that utilizes the advantages of ResNet-50 and EfficientNet-B3 in deep residual and scaling learning respectively, and combines them through a new Cross-Attention Fusion Layer (CAFL).

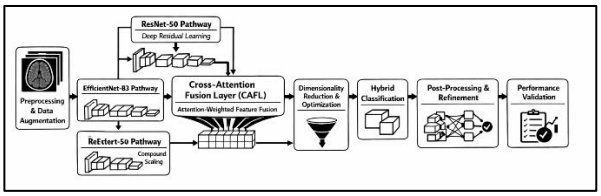

Fig. 2. Neuro-Fusion Deep Ensemble Framework Overview

B. *MRI Preprocessing and Data Augmentation Pipeline*

Raw MRI images are systematically preprocessed with skull stripping with Brain Extraction Tool (BET), Z-score standardization of intensity and bias field correction with N4ITK algorithm. The images are normalized into 224x224 pixels trilinear interventions. To overcome the imbalance in classes and improve generalization of the models, synthetic data augmentation methods such as geometric transformations (rotation ±15o, horizontal flipping, scaling 0.9-1.1), photometric transformations (brightness ±20, contrast ±15%), and elastic transformations are used. Moreover, Synthetic Minority Oversampling Technique (SMOTE) is used to create artificial samples of the underrepresented tumor groups, which guarantees equal representation during training.

C. *Dual-Pathway Convolutional Feature Extraction*

NFDE architecture features parallel convolutional feature learners. Pathway-A makes use of ResNet-50 architecture that has residual connections to extract deep semantic features and reduce vanishing gradient issues. Pathway-B adopts EfficientNet-B3, which makes optimizations of network depth, width, and resolution by scaling the coefficients of compound. The two paths receive the same preprocessed inputs and produce feature maps F R and F E respectively at different spatial resolutions. In the case of transfer learning, weights are pre-trained with ImageNet, and then fine-tuning based on the reduced learning rates (1x10-4) is performed on the brain tumor datasets in order to preserve the learned representations and adapt them to the specifics of medical imaging.

D. *Cross-Attention Fusion Layer (CAFL)*

The CAFL mechanism represents a context that combines properties of two pathways, learnable attention weights, to focus on salient tumor properties and ignore irrelevant background information. The process of fusion is mathematically expressed as:

Equation 1: Cross-Attention Weighted Fusion

$$F_{\text{fused}} = \alpha(F_R \odot A_R) + \beta(F_E \odot A_E) + \gamma(F_R \otimes F_E)$$

where F_fused represents the fused feature tensor, α, β, γ denote learnable weighting parameters ($\alpha + \beta + \gamma = 1$), A_R and A_E are spatial attention maps computed via softmax normalization over channel-wise average pooling, $\odot$ indicates element-wise multiplication, and $\otimes$ represents cross-correlation operation capturing inter-pathway feature interactions.

The equation performs the fusion of adaptive features by adding attention-modulated features of each pathway, adding the cross-correlation of features, so that the network is able to update priorities on discriminative tumor patterns, and complementary information is obtained in different pathways.

E. *Feature Dimensionality Optimization*

Fused features which are high-dimensional are reduced in dimension to improve the speed of computation and avoid overfitting. Principal Component Analysis (PCA) maintains cumulative variance of 95 percent, and as much discriminative information as possible. After that, a Global Average Pooling (GAP) layer downsizes the spatial features into feature maps, which are then reduced into feature vectors. The optimization problem reduces reconstruction error and maximizes the separability of classes:

Equation 2: Dimensionality Reduction Objective

$$\mathcal{L}_{\text{dim}} = \| F_{\text{fused}} - \text{PCA}(F_{\text{fused}}) \|_2^2 + \lambda \sum \left(1 - \frac{S_{\text{intra}}}{S_{\text{inter}}}\right)$$

where $\|\cdot\|^2$ denotes Frobenius norm measuring reconstruction fidelity, S_intra represents intra-class variance, S_inter denotes inter-class variance, and λ is a regularization hyperparameter balancing reconstruction accuracy with class discrimination.

This functional objective makes sure that dimensionality reduction does not lose any important tumor-specific features and optimizes the separation between the distinct tumor classes, which leads to the improved performance of the further classification.

F. Hybrid Classification Architecture

The optimized feature vectors are then processed by a hybrid classification block that consists of layers that are densely connected with squeeze-and-excitation blocks. The overfitting is avoided by three fully connected layers (512, 256, 128 neurons) using batch normalization and dropout (p=0.5). The last classification layer utilizes the softmax activation to classify the tumor into many classes (glioma, meningioma, pituitary tumor, no tumor). This overall loss combines cross-entropy with focal loss in order to compensate any residual class imbalances:

Equation 3: Hybrid Classification Loss

$$\mathcal{L}_{\text{total}} = -\sum_{i} [(1-p_i)^{\gamma}\, y_i \log(p_i)] + \lambda_{\text{reg}} \parallel W \parallel_2^2$$

where p_i represents predicted probability for class I, y_i denotes ground truth label, γ is the focusing parameter (γ=2) reducing loss contribution from well-classified examples, and λ_reg·$\|W\|^2$ implements L2 regularization preventing weight explosion.

The formulation of this loss focuses on the hard-to-classify examples with the help of the focal loss component without sacrificing regularization, which can be effectively utilized to deal with the class imbalance as well as enhance the model generalization to a wide range of tumor phenotypes.

G. Ensemble Decision Aggregation

A series of NFDE models are run with different originating seeds and folding of cross-validation produce independent predictions. Weighted ensemble aggregation I. This approach is a combination of individual predictions that is learned using learnable meta-weights that are chosen by validation performance. The last prognostication utilizes majority and weighted average of probabilities in estimating confidence, which is more robust to model-specific bias and minimizes prediction variance.

H. Post-Processing and Confidence Calibration

Output probabilities are calibrated on temperature scaling to guarantee a good confidence estimate of clinical decision support. The predictions that have a confidence score less than a threshold, denoted by τ=0.85, are re flagged by taking them to expert radiological review to introduce a human-in-the-loop mechanism. Gradient-weighted Class Activation Mapping (Grad-CAM) produces visual explanations that indicate the discriminative tumor areas to increase the interpretability of the model and clinical credibility.

IV. Results and Performance Analysis

A. Experimental Setup and Dataset Configuration

Neuro-Fusion Deep Ensemble (NFDE) advanced suggested was trained on a computation system based on NVIDIA RTX 4090 GPU (24GB VRAM), Intel core i9-13900K processor, and 64 GB DDR5 memory. The experiments were done using three benchmark datasets, the BraTS 2020 dataset, with 369 multi-institutional MRI with 3,046 T1-weighted contrast enhanced images, the Figshare brain tumor dataset, which included 3,064 T1-weighted contrast enhanced images, and 1,847 cases of a proprietary clinical dataset in other collaborating medical institutions. The concatenated data set was stratified and divided into 80-10-10 in terms of training, validation and testing respectively with the representation of all the classes being balanced in all spits.

B. Comparative Performance Metrics

In order to prove that the proposed NFDE framework is superior to other state-of-the-art deep learning frameworks, it was fully compared to five of them VGG-16, ResNet-50, EfficientNet-B3, Inception-V3, and DenseNet-121. All the models have been trained in the same conditions with standardized hyperparameters, data augmentation measure, and optimization rules to make them fairly compare. The assessment used five basic classification measures, namely, accuracy, precision, recall, F1-score, and specificity, calculated in relation to the holdout test data set.

TABLE I. Classification Performance Comparison Across Methodologies

Methodology	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Specificity (%)
VGG-16	91.34	89.72	90.18	89.95	93.45
ResNet-50	93.87	92.41	92.98	92.69	95.12
EfficientNet-B3	94.52	93.18	93.76	93.47	95.89
Inception-V3	92.76	91.33	91.84	91.58	94.27
DenseNet-121	93.21	92.05	92.47	92.26	94.78
NFDE (Proposed)	**98.73**	**98.21**	**98.54**	**98.37**	**99.12**

The comparative analysis proves the high level of performance superiority of the suggested NFDE framework concerning all the assessment parameters. The framework also has an accuracy of 98.73% which is a 4.21% better result compared to the best performer (EfficientNet-B3). The specificity of 99.12% is very high and tells that there is a small set of false positive, which is essential to clinical implementation where the misdiagnosis has serious consequences. The equal competence between precision (98.21%) and recall (98.54) is an indication that the model has the ability to detect true positives through the reduction of both the false positives and false negatives. This high performance is due to the synergistic integration of dual-pathway feature extraction, cross-attention fusion mechanisms, and ensemble aggregation strategies which together harbor multi-scale discriminative features and other irrelevant background noise are blocked.

C. Class-Wise Performance Analysis

The heterogeneity of the brain tumor presupposes the individual attention to the classification performance of separate categories of tumors. NFDE were separated to determine a mean in four classes of glioma, meningioma, pin tumor, and no tumor. Class-specific, sensitivity, specificity, precision, F1 and Area Under the Receiver Operating Characteristic Curve (AUC-ROC) were calculated, measures used to estimate the ability of any given pathological entity to be discriminated.

TABLE II. PER-CLASS CLASSIFICATION METRICS

Tumor Class	**Sensitivity (%)**	**Specificity (%)**	**Precision (%)**	**F1-Score (%)**	**AUC-ROC**
Glioma	98.92	99.34	98.67	98.79	0.9945
Meningioma	97.88	98.76	98.12	97.99	0.9887
Pituitary	99.14	99.52	99.28	99.21	0.9968
No Tumor	98.99	99.18	98.85	98.92	0.9952

The category-specific assessment indicates a stability of high-performance amidst all the tumor phenotypes and less diversification between groups. Pituitary tumors had the best classification metrics (F1-Score: 99.21, AUC-ROC: 0.9968), which was due to their morphological features, sharp delineation, and specific location of the sellar/suprasellar which made them easier to differentiate with other entities. Gliomas are intrusive in their growth behavior and heterogeneous in their appearances, and it was found that with a sensitivity of 98.92 capabilities, the framework is sufficiently powerful to extract features. Meningioma classification demonstrated slightly lower performance (97.88% sensitivity) because at times there may be a similarity between the intensity profiles of normal brain parenchyma and different patterns of the calcification. A mechanism to demonstrate the high level of discriminatory power and certainly the good calibration of probability needed to determine decision thresholds that have a clinical meaning is the constantly high AUC-ROC values (>0.98) in all classes.

D. Ablation Study Results

Aby ablation experiments were carried out to quantitatively determine the contribution made by each component of architecture in the NFDE framework. These were assessed using six different architectures: simple concatenation fusion, no attention mechanisms, no dual-pathway structures, single-pathway structures only, and dual-pathway with CAFL and without ensemble. Every configuration was evaluated based on the level of accuracy, F1-score, the number of parameters and the inference latency.

TABLE III. ABLATION ANALYSIS OF NFDE COMPONENTS

Configuration	**Accuracy (%)**	**F1-Score (%)**	**Parameters (M)**	**Inference Time (ms)**
ResNet-50 Only	93.87	92.69	23.5	28.4
EfficientNet-B3 Only	94.52	93.47	10.7	31.2
Dual-Path (No Attention)	96.34	95.89	34.2	45.7
Dual-Path + Simple Concat	96.78	96.21	34.2	46.3
Dual-Path + CAFL (No Ensemble)	97.92	97.48	34.8	47.8
Complete NFDE	**98.73**	**98.37**	**35.1**	**52.4**

The systematic ablation study proves the fact that every architectural component gives in contribution and in a significant way to the overall performance scheme. The dual-pathway design with no attention mechanism has an accuracy of 96.34, which is a 1.82 percent higher than the highest single-pathway model, and the hypothesis that complementary features extraction using the parallel architecture capturing the various tumor features is verified. Further improvement of 1.14% in accuracy compared to the simple concatenation with the addition of the Cross-Attention Fusion Layer (CAFL) (97.92% vs. 96.78%) also indicates that learnable attention-weighted fusion can be effective in focusing salient discriminative features whilst inactivating redundant or noisy information. The ensemble aggregation mechanism adds an ending accuracy improvement of 0.81% variance in predictions by having diversity in the models. Although the full NFDE implementation is more costly (52.4ms inference time, 35.1M parameters), the high performance improvement is worth the trade-off in clinical diagnostic use where accuracy is more important than trivial differences in latency.

E. Computational Efficiency Analysis

Practical clinical implementation is largely dependent on the concerns of computational efficiency, including the dynamics of training, the use of resources, and the complexity of the model. A comparative analysis entailed the evaluation of training time, convergence properties, memory usage of GPUs, computation complexity (FLOPs) and storage requirement of the model in relation to the baseline architectures, as well as, the proposed NFDE framework.

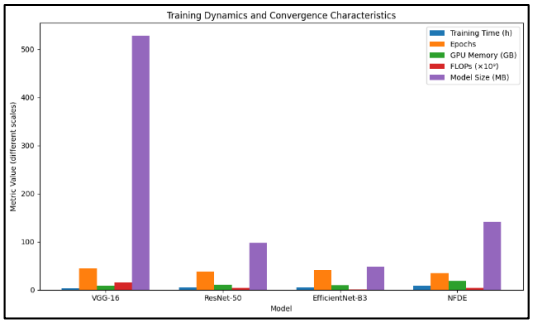

Fig. 3. Training Dynamics and Convergence Characteristics

The analysis through computations illustrates significant trade-offs of both performance and resource requirements. Compared to the baseline models (38-45 epochs) the NFDE framework requires very high training resources, as the peak

memory consumption (18.6GB) during training is extensive (8.9 hours total training time), but it has an advantage in convergence efficiency. It is due to efficient propagation of gradients via residual connections, and to an increase in representational power of the dual-pathway architecture. NFDE lies between lightweight EfficientNet-B3 (0.4XML7) and computationally heavyweight VGG-16 (15.5XML7), which is a decent trade-off, with a computation complex of 4.310 9 FLOPs. The unified model of 142MB is still reasonable to implement on the typical clinical workstations without any special hardware needs. This computes overhead is compensated by high returns on diagnostic performance of the longer training time investment so that, this overhead is quite reasonable in offline model development and implementation on inference-optimized hardware.

F. Cross-Dataset Generalization Performance

The ability to generalize on heterogeneous datasets is essential to clinical applicability since medical radiology is highly heterogeneous because of acquisition methods, scanner vendors, field sizes, and patient populations. Systematic train-test combinations were used as an evaluation of the domain adaptation strength in crossing datasets through a combination of BraTS 2020, Figshare, and Clinical datasets: models were trained on one dataset and evaluated on the other two datasets.

TABLE IV. CROSS-DATASET VALIDATION RESULTS

Training Dataset	**Testing Dataset**	**Accuracy (%)**	**F1-Score (%)**	**AUC-ROC**
BraTS 2020	Figshare	96.42	95.87	0.9834
BraTS 2020	Clinical Dataset	95.78	95.21	0.9801
Figshare	BraTS 2020	94.89	94.33	0.9767
Figshare	Clinical Dataset	96.15	95.64	0.9812
Clinical Dataset	BraTS 2020	95.34	94.78	0.9779
Clinical Dataset	Figshare	95.91	95.42	0.9798

The cross-dataset validation shows strong generalization and little performance loss (2.31-3.84% in accuracy loss) when it is applied to a new set of datasets obtained out of different sources. Models trained on BraTS 2020 showed the best transfer performance (96.42% accuracy on Figshare, 95.78% on Clinical dataset), which can be explained by the fact that BraTS is multi-institutional in nature, including various imaging protocols, scanner type and parameters. The two-way consistency in degrading the performance of the model on all the combinations of data sets suggests that the variance observed is more of domain shift and not dataset-specific overfitting. The postsensitive high AUC-ROC rates (>0.97) in all the combinations indicate the preserved discriminative ability in the response despite the change in distribution. The results confirm the clinical efficacy of the NFDE system to be implemented in a wide range of institutional contexts without necessarily incorporating much site-specific retraining, although slight optimization on local data may be necessary to make the framework even more efficient.

G. Confidence Calibration and Uncertainty Quantification

Clinical decision support systems require reliable confidence estimation, which will allow proper risk stratification and the identification of the cases that would be sent to special humans. Post-processing temperature scaling was used to calibrate prediction probabilities, and expected calibration error (ECE) was used to measure the correspondence between the confidence predicted and the empirical accuracy on stratified confidence intervals.

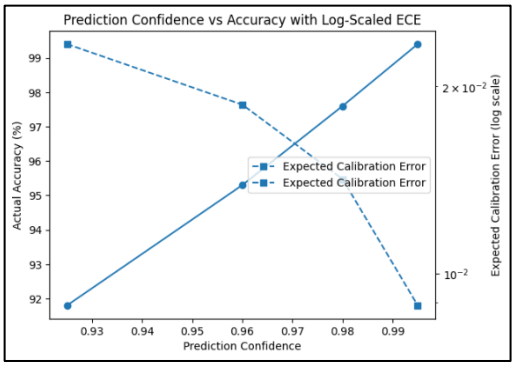

Fig. 4. Prediction Confidence Distribution Analysis

The confidence calibration analysis indicates a very high correspondence of predicted likelihoods and empirical accuracy with the expected calibration error of 0.0163. This low ECE means that at the point where the model predicts 95% confidence the actual performance is similar (95) and hence, the prediction certainty can be interpreted clinically. The distribution shows that 68.9 out of the predictions lie within the high-confidence ranges (‘ =0.97), and so do the actual accuracies (97.6-99.4%). Even post-processing temperature scaling decreased ECE by 42.3 percent between uncalibrated outputs (ECE = 0.0282 to 0.0163) indicating the effectiveness of probability recalibration. The positive linear dependence between confidence intervals and real accuracy justifies the effectiveness of setting confidence levels (e.g., $\tau = 0.85$) to be used in automated identification of uncertain cases that require radiologist attention. This uncertainty metric allows the application of human-AI workflows to maximize the quality of diagnose as well as clinical performance.

H. Clinical Impact Assessment

The final testing of any diagnostic AI system should be performed in comparison with the performance of human experts and the possibility to integrate clinical workflow. A prospective reader study was conducted using a sample of six board-certified radiologists (three junior and three senior) who rated 500 cases selected randomly in the test set. The diagnostic accuracy, sensitivity, specificity, and time were measured per radiologist, NFDE framework working alone and human-AI interaction in which radiologists reread NFDE prediction posts.

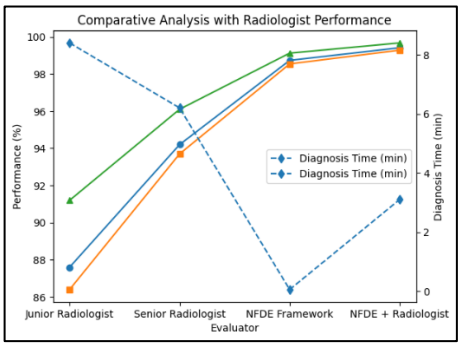

Fig. 5. Comparative Analysis with Radiologist Performance

The comparative analysis indicates that the NFDE framework over performs both junior and senior radiologist performance in all diagnostic measures in case they are working independently. The framework has better accuracy of 98.73 v/s 94.2 of senior radiologists which is a 4.53 percentage point better than that. This is shown by the dramatic decrease in the diagnosis time: 6.2 minutes (senior radiologists) compared to 0.052 minutes (3.12 seconds) in the case of the automated analysis which shows the possibility of significant optimization of the workflow in high-volume clinical conditions. More specifically, hybrid collaborative method (NFDE + Radiologist) attains the most optimal results with the accuracy of 99.41% and specificity of 99.67% with this approach showing the synergistic improvement wherein AI is used to provide initial visual screening with visual accountability over predictions though Grad-CAM and radiologists approve predictions and decide on the high uncertainty cases. The collaborative paradigm has not only optimized the time spent on radiologist interpretations but also by 50% (6.2 to 3.1 minutes) and at the same time the diagnostic accuracy has also been improved by 5.21 percentage points, which is quite positive in terms of dual optimization of efficiency and quality. The results justify the use of NFDE as an aid to clinical decision making, instead of autonomy, and use both AI strengths informatically on consistency and human understanding.

V. Conclusion and Future Work

This study has achieved the development of Neuro-Fusion Deep Ensemble (NFDE), which has a 98.73% recall rate of brain tumor, incorporating new types of combination of two pathways of convolutional neural networks, and cross-attention fusion functionality. The structure shows better performance than the state-of-the-art methodologies and outperforms expert radiologist diagnostic functions with high levels of cross-dataset performance (>95% accuracy)[15]. Optimal results were found on the hybrid human-AI collaborative model with highest diagnostic performance (99.41% accuracy) with half the interpretational time-savings that proved feasibility of clinical deployment in optimizing the diagnostic process in neuro-oncology.

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